

set_wifi

REFERENCE

API consumer

create or modify a Wi-Fi profile on the reader

Overview

`set_wifi` writes a Wi-Fi profile to the reader — it **creates** a new saved profile or **modifies** an existing one (selected by `wifiConfig.operation` = `CREATE` / `MODIFY`). One payload configures the interface, the access point (ESSID, preference), the security type and its credentials (WPA2/WPA3 Personal or Enterprise), and DHCP IPv4. It is a state-changing command on the device-management (MGMT) endpoint. The response is status-only — it echoes **no configuration and no credentials**; confirm the result with `get_wifi`.

`set_wifi` **is state-changing**: `accessPoint.connect: true` **switches networks and can sever the MQTT control session**. Use `connect: false` to save a profile without switching. See the warning callout in [Notes and Caveats](#).

Details

Property	Value
Interaction Pattern	Request / Response (app publishes a command; device publishes the correlated reply)
State Change	STATE-CHANGING (writes a saved profile; may switch the active connection)
Reversibility	REVERSIBLE (remove a created profile with <code>delete_wifi_profile</code> ; re-send <code>set_wifi</code> <code>MODIFY</code> to change settings)
Supported Operations	<code>CREATE</code> (new profile), <code>MODIFY</code> (update existing, matched by <code>ssid</code>)
Required Request Fields	<code>command</code> , <code>requestId</code> , <code>wifiConfig</code> (with <code>operation</code> and <code>accessPoint</code> ; inside it, <code>ssid</code> and <code>isPreferred</code>)
Applies To	Endpoint: Management (MGMT). Reader: RFD40 series, RFD90 series
API Version	<code>v1.1</code> (API Version Reporting)
Related	<code>get_wifi</code> , <code>delete_wifi_profile</code> , <code>install_certificate</code>

When to Use

Use `set_wifi` to **provision or update the reader's Wi-Fi** — during onboarding, when a network changes, or to mark a profile preferred.

- `CREATE`: first-time provisioning of a new ESSID. Fails with code `18` if the ESSID already exists.
- `MODIFY`: update an existing profile (password, security, preference), matched by `ssid`. Fails with code `15` if the ESSID is not found. A `MODIFY` can be a partial payload — e.g. just `operation` + `accessPoint.{isPreferred, ssid}` to set the preferred flag.
- Pre-provision safely**: send `connect: false` to save a profile without switching the active connection — the recommended pattern when managing a reader over the same Wi-Fi that carries the MDM session.

Topic Mapping

Message	Direction	Topic	Note
Request	App → Reader	<code><tenantId>/MGMT/clients/cmnd/<serial></code>	On the management endpoint.
Response	Reader → App	<code><tenantId>/MGMT/clients/resp/<serial></code>	Correlated reply to the request above; the in-payload <code>requestId</code> links the twin topics.

Topics follow the pattern `<tenantId>/<topic>/<serial>` — the tenant id assigned to you at provisioning (a slot you fill in, never a literal), the base topic you configure on the endpoint, and the reader's serial number, which the reader appends itself in its model-prefixed form (for example `RFD40-212735201D0053`). `App → Reader` means your app publishes and the reader subscribes; `Reader → App` means the reader publishes and your app subscribes.

Credentials are presented once at `CONNECT`. The `requestId` you send is echoed in the response so you can pair reply to request. The endpoint's `mqttParams` govern QoS and retain, not this command.

See [Topics and Channels](#).

Key Fields

- `wifiConfig.operation` (request) — `CREATE` or `MODIFY`; determines whether a new profile is added or an existing one (matched by `ssid`) is updated.

- `wifiConfig.accessPoint.essid` (request) — the network name; the create target / modify key (case-sensitive).
- `wifiConfig.accessPoint.connect` (request) — ⚠️ `true` switches the active connection (can sever the MDM session); `false` just saves. See the notes below.
- `wifiConfig.accessPoint.security.securityType` (request) — selects which `securityDetails` key you populate (WPA2/WPA3 Personal or Enterprise).
- `response.code` (response) — read this to determine success vs failure (see [Response Codes](#)).

Request Payload

The request envelope is `command`, `requestId`, and the `wifiConfig` payload. Use the **Schema** tab for the full field reference and the **Example** tab for a ready-to-send WPA2-Personal `CREATE` (enterprise variants are in [Examples](#)).

Credentials in the examples are the contract's sample values, not real secrets — passwords, passphrases, and identities are write-only (see [Request Semantics](#)).

Pick a **profile scenario** below; the Example and Schema update together. Use the **Schema** tab for the full field reference and the **Example** tab for the concrete payload.

Profile scenario

Create WPA2 personal



Request payload

Example: WPA2-Personal CREATE

```
1  {
2    "command" : "set_wifi" ,
3    "requestId" : "abc123" ,
4    "wifiConfig" : {
5      "interfaceName" : "wlan0" ,
6      "enableInterface" : true ,
7      "operation" : "CREATE" ,
8      "accessPoint" : {
9        "connect" : false ,
10       "isPreferred" : false ,
11       "autoConn" : true ,
12       "essid" : "TestAP1" ,
13       "enableSecurity" : true ,
14       "security" : {
15         "securityType" : "WPA2Personal" ,
16         "securityDetails" : {
17           "WPA2Personal" : {
18             "password" : "test@123"
19           }
20         }
21       }
22     }
23   }
24 }
```

Request payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE
1	command *	string	—	set_v
2	requestId *	string	—	abc12
3	wifiConfig *	object	—	—
4	interfaceName	string	—	wlan0

#	FIELD	TYPE	ENUM / UNITS	EXAMP
5	enableInterfac e	boolean	—	true
6	operation *	enum	CREATE MODIFY	CREAT
7	accessPoint *	object	—	—
8	connect	boolean	—	false
9	isPreferre d *	boolean	—	false
10	autoConn	boolean	—	true
11	essid *	string	—	ZWire
12	enableSecur ity	boolean	—	true

#	FIELD	TYPE	ENUM / UNITS	EXAMP
13	ipv4Configu ration	object	—	—
14	ipAddress	string	ipv4	192.1
15	subnetMas k	string	ipv4	255.2
16	gateway	string	ipv4	192.1
17	enableDhc p	boolean	—	true
18	dnsServer	string	ipv4	192.1
19	domainNam e	string	—	exam
20	security	object	—	—
21	securit yType *	enum	WPA2Personal WPA3Personal WPA2Enterprise WPA3Enterprise	WPA2P
22	securit yDetail * s	object	—	—
23	WPA2Pe rsonal	object	—	—

#	FIELD	TYPE	ENUM / UNITS	EXAMP
24	password	string	—	test()
25	WPA3Personal	object	—	—
26	password	string	—	—
27	WPA2Enterprise	object	—	—
28	authentication	enum	tls ttls peap	peap
29	innerAuthentication	enum	none mschapv2	mscha
30	identity	string	—	test
31	anonymousIdentity	string	—	test

#	FIELD	TYPE	ENUM / UNITS	EXAMP
32	password	string	—	test0
33	password	string	—	test0
34	certificate	array	—	—
35	key	enum	ca_cert client_key client_cert cert_key_password	ca_ce
36	value	string	—	wifi_
37	protocol	enum	WPA2_Enterprise_CCMP	WPA2_
38	WPA3Enterprise	object	—	—
39	authentication *	enum	tls ttls peap	peap

#	FIELD	TYPE	ENUM / UNITS	EXAMP
40	innerAuthentication	enum	none mschapv2	mscha
41	identity	string	—	test
42	anonymousIdentity	string	—	test
43	password	string	—	test
44	phrase	string	—	test
45	certificate	array	—	—
46	key	enum	ca_cert client_key client_cert cert_key_password	ca_ce
47	value	string	—	wifi_

#	FIELD	TYPE	ENUM / UNITS	EXAMP
48	profile	enum	WPA3_Enterprise_CCMP WPA3_Enterprise_CCMP_256 WPA3_Enterprise_GCMP_128 WPA3_Enterprise_GCMP_256_SHA256 WPA3_Enterprise_GCMP_256_SUITEB_192	WPA3_

Request Semantics

- **CREATE vs MODIFY.** `CREATE` adds a new profile (fails `18` if the `ssid` exists); `MODIFY` updates an existing profile matched by `ssid` (fails `15` if not found). The device enforces a maximum number of saved profiles — exceeding it returns `19`; delete unused profiles with `delete_wifi_profile`. For Enterprise TLS, the `ca_cert` / `client_cert` / `client_key` referenced in `certificate` must already be installed via `install_certificate`.
- **Credentials are write-only — mask them.** Supply passwords, passphrases, and identities in the request (the examples carry sample values); the device never echoes them — `set_wifi`'s response carries no credentials, and `get_wifi` returns stored secrets masked.
- **Enterprise inner authentication.** For `ttls` and `peap`, `innerAuthentication` must be `mschapv2` and defaults to `mschapv2` when omitted; for `tls` it is not applicable and can be omitted, as the `tls` variants on this page do.
- **IPv4 only, DHCP only.** In the current API version, `set_wifi` supports **IPv4 with DHCP assignment only**. The `ipv4Configuration` static fields (`ipAddress` / `subnetMask` / `gateway` / `dnsServer`) are modeled in the schema but **not supported yet** (planned for a future release), and **IPv6 is not available**. Use `enableDhcp: true`.

Response Payload

`set_wifi` returns a **status-only** envelope — `command`, `requestId`, `apiVersion`, and `response`. There is no configuration echo and no credential in the response; to confirm the saved profile, follow up with `get_wifi`.

Response payload

```

1  {
2    "command" : "set_wifi" ,
3    "requestId" : "abc123" ,
4    "apiVersion" : "V1.1" ,
5    "response" : {
6      "code" : 0 ,
7      "description" : "Success"
8    }
9  }

```

Response payload

* required • — not applicable • Colored badge = JSON type

#	FIELD	TYPE	ENUM / UNITS	EXAMPLE	DESCRIPTION
1	command *	string	—	set_wifi	Echoes the executed command.
2	requestId *	string	—	abc123	Echoes the request's requestId — pairs reply to request.
3	apiVersion *	enum	V1.0 V1.1 V1.2 V1.21 V1.22	V1.1	API version reported by this command. Emitted per-handler, not device-wide.
4	response *	object	—	—	Status-only result — set_wifi echoes no configuration and no credentials.
5	code *	integer	0-101	0	0 = Success; non-zero = error. Required by the contract. See Response Codes below.
6	description *	string	—	Success	Human-readable status.

Response Codes

`set_wifi` returns `response.code`. The codes applicable to this command are below; for the full table see the [Response Codes](#) and [Correlation and Request/Response Pairing](#).

Code	Description
0	Success — the profile was created or modified.
2	Invalid payload (<code>IOT_ERROR_INVALID_PAYLOAD</code>) — the JSON is malformed or has invalid field values. Validate against the schema and correct.
15	WIFI Error - SSID not found (<code>IOT_ERROR_SSID_NOT_FOUND</code>) — a <code>MODIFY</code> targeted an <code>essid</code> that is not saved. Verify the name or <code>CREATE</code> it first.
17	WIFI Error - SSID missed (<code>IOT_ERROR_SSID_MISSED</code>) — the required <code>essid</code> field is missing from the payload.
18	WIFI Error - SSID already exist (<code>IOT_ERROR_SSID_ALREADY_EXIST</code>) — a <code>CREATE</code> used an <code>essid</code> that already exists. Use <code>MODIFY</code> , or delete the existing profile first.
19	WIFI Error - SSID count overflow (<code>IOT_ERROR_SSID_LIMIT_OVERFLOW</code>) — the saved-profile limit is reached. Delete an unused profile with <code>delete_wifi_profile</code> .
20	Wifi is not supported (<code>IOT_ERROR_WIFI_INTER_NOT_SUPPORTED</code>) — the device has no Wi-Fi interface; use Ethernet.
23	Invalid enum value (<code>IOT_ERROR_INVALID_ENUM</code>) — a field (e.g. <code>operation</code> , <code>securityType</code> , <code>authentication</code>) is outside its allowed set.
101	Default error — generic catch-all.

Examples

Publish a `set_wifi` request; the device replies with the status-only envelope shown under [Response Payload](#). Every contract example is in the **Profile scenario** dropdown under [Request Payload](#):

- **Create WPA2 personal and Modify WPA2 personal** — a pre-shared key.
- **Set preferred AP** — a partial `MODIFY` that needs only `operation` plus `accessPoint`.
`{isPreferred, essid}`.
- **Create WPA2 enterprise TLS / TTLS / PEAP and Create WPA3 enterprise TLS / TTLS / PEAP** — installed certificates referenced as `{key, value}`, an EAP `authentication` method and a `protocol`; the TTLS and PEAP variants add `innerAuthentication`.

Notes and Caveats



`connect: true` **CAN SEVER THE CONTROL SESSION**

`accessPoint.connect: true` disconnects the currently connected profile and connects to the target ESSID. If the reader manages itself over the same Wi-Fi that carries its MQTT/MDM session, switching to a different network **drops the control channel** and can strand the device. When managing a reader over Wi-Fi, use `connect: false` (save without switching), and undo an unwanted profile with `delete_wifi_profile` rather than connecting onto it.

Because the response is status-only, **verify the result with** `get_wifi` — check that the profile was saved (and its `isPreferred` / `accessPoint.status`). For the concept-level view, see Configure Wi-Fi.

Related: [Configure Wi-Fi](#) · `get_wifi` · `delete_wifi_profile` · `install_certificate` · [Response Codes](#)